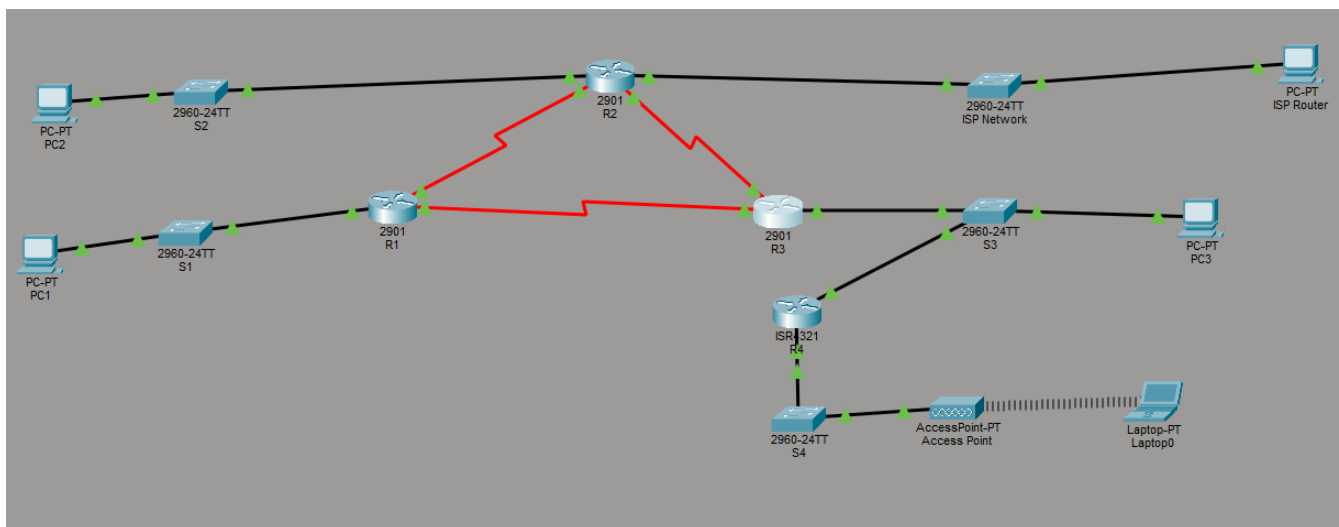


รายงานผล Lab12

นายจิรภัทร แก้วดี 673380577-9 Sec 2

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	G0/1	64.100.54.6	255.255.255.252	
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	G0/1	192.168.11.1	255.255.255.0	
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
R4	G0/0/0	192.168.1.2	255.255.255.0	N/A
	G0/0/1	192.168.11.1	255.255.255.0	
ISP Router	NIC	64.100.54.5	255.255.255.252	N/A
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Laptop	NIC	DHCP	DHCP	DHCP



```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#username BranchAdmin privilege 15 secret Branch1234
R1(config)#username Admin privilege 15 secret Cisco1234
R1(config)#line vty 0 4
R1(config-line)# login local
R1(config-line)# exit
R1(config)#interface g0/0
R1(config-if)# ip address 172.16.1.1 255.255.255.0
R1(config-if)# no shutdown

R1(config-if)# exit
R1(config)#interface g0/1
R1(config-if)# ip address 64.100.54.6 255.255.255.252
R1(config-if)# no shutdown

R1(config-if)# exit
R1(config)#interface s0/0/0
R1(config-if)# ip address 172.16.3.1 255.255.255.252
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)# exit
R1(config)#interface s0/0/1
R1(config-if)# ip address 192.168.10.5 255.255.255.252
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R1(config-if)# exit
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 172.16.1.0 0.0.0.255 area 0
R1(config-router)# network 172.16.3.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.4 0.0.0.3 area 0
R1(config-router)# passive-interface g0/0
R1(config-router)# end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

R1#
%SYS-5-CONFIG_I: Configured from console by console

```

แสดงการตั้งค่า R1 ผ่าน CLI ได้แก่ hostname, username สำหรับ login, IP address ทุก interface (G0/0, G0/1, S0/0/0, S0/0/1) และการตั้งค่า OSPF area 0 พร้อม passive-interface บน G0/0

```

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#username BranchAdmin privilege 15 secret Branch1234
R2(config)#username Admin privilege 15 secret Cisco1234
R2(config)#line vty 0 4
R2(config-line)# login local
R2(config-line)# exit
R2(config)#interface g0/0
R2(config-if)# ip address 172.16.2.1 255.255.255.0
R2(config-if)# no shutdown

R2(config-if)# exit
R2(config)#interface s0/0/0
R2(config-if)# ip address 172.16.3.2 255.255.255.252
R2(config-if)# clock rate 128000
R2(config-if)# no shutdown

R2(config-if)# exit
R2(config)#interface s0/0/1
R2(config-if)# ip address 192.168.10.9 255.255.255.252
R2(config-if)# clock rate 128000
This command applies only to DCE interfaces
R2(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R2(config-if)# exit
R2(config)#ip route 0.0.0.0 0.0.0.0 172.16.3.1
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R2(config)#router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 172.16.2.0 0.0.0.255 area 0
R2(config-router)# network 172.16.3.0 0.0.0.3 area 0
R2(config-router)# network 192.168.10.8 0.0.0.3 area 0
R2(config-router)# passive-interface g0/0
R2(config-router)# default-information originate
R2(config-router)# end
R2#
%SYS-5-CONFIG_I: Configured from console by console

00:03:14: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/0/0 from LOADING to FULL, Loading
Done

```

แสดงการตั้งค่า R2 รวมถึง IP address, clock rate บนพอร์ต Serial ฝั่ง DCE, static default route ไปยัง ISP (172.16.3.1), และ OSPF พร้อมคำสั่ง default-information originate เพื่อประกาศ default route เข้าสู่ระบบ OSPF ผลลัพธ์แสดง Adjacency กับ R1 (1.1.1.1) สำเร็จเป็น FULL

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface g0/0
R3(config-if)# ip address 192.168.1.1 255.255.255.0
R3(config-if)# no shutdown

R3(config-if)# exit
R3(config)#interface s0/0/0
R3(config-if)# ip address 192.168.10.6 255.255.255.252
R3(config-if)# no shutdown

R3(config-if)# exit
R3(config)#interface s0/0/1
R3(config-if)# ip address 192.168.10.10 255.255.255.252
R3(config-if)# no shutdown

R3(config-if)# exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

R3(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R3(config)#router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 192.168.1.0 0.0.0.255 area 0
R3(config-router)# network 192.168.10.4 0.0.0.3 area 0
R3(config-router)# network 192.168.10.8 0.0.0.3 area 0
R3(config-router)# passive-interface g0/0
R3(config-router)# end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#
00:03:46: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0/1 from LOADING to FULL, Loading
Done

00:03:46: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/0/0 from LOADING to FULL, Loading
Done

```

แสดงการตั้งค่า R3 ได้แก่ IP ทุก interface และ OSPF area 0 ผลลัพธ์แสดง Adjacency กับทั้ง R1 (1.1.1.1) และ R2 (2.2.2.2) สำเร็จเป็น FULL ทั้งคู่

```

Router>configure terminal
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#interface g0/0/0
R4(config-if)# ip address 192.168.1.2 255.255.255.0
R4(config-if)# no shutdown

R4(config-if)# exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R4(config)#interface g0/0/1
R4(config-if)# ip address 192.168.11.1 255.255.255.0
R4(config-if)# no shutdown

R4(config-if)# exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down

R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 192.168.1.0 0.0.0.255 area 0
R4(config-router)# network 192.168.11.0 0.0.0.255 area 0
R4(config-router)# passive-interface g0/0/1
R4(config-router)# exit
R4(config)#ip dhcp excluded-address 192.168.11.1
R4(config)#ip dhcp pool BRANCH-LAN
R4(dhcp-config)# network 192.168.11.0 255.255.255.0
R4(dhcp-config)# default-router 192.168.11.1
R4(dhcp-config)# exit
R4(config)#end
R4#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R4#
R4#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	192.168.1.2	YES	manual	up	up
GigabitEthernet0/0/1	192.168.11.1	YES	manual	down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

R4#

```

แสดงการตั้งค่า R4 ได้แก่ hostname, IP บน G0/0/0 และ G0/0/1, OSPF area 0 พร้อม passive-interface บน G0/0/1 (ฝั่ง Branch LAN), และการตั้งค่า DHCP pool สำหรับแจก IP ให้อุปกรณ์ในเครือข่าย 192.168.11.0/24

Objectives

In this lab, you will use the CLI commands to verify the operation of an existing OSPFv2 network. In Part 2, you will add a new LAN to the configuration and verify connectivity.

- Identify and verify the status of OSPF neighbors.
- Determine how the routes are being learned in the network.
- Explain how the neighbor state is determined.
- Examine the settings for the OSPF process ID.
- Add a new LAN into an existing OSPF network and verify connectivity.

Background / Scenario

You are the network administrator for a branch office of a larger organization. Your branch is adding a new wireless network into an existing branch office LAN. The existing network is configured to exchange routes

using OSPFv2 in a single-area configuration. Your task is to verify the operation of the existing OSPFv2 network, before adding in the new LAN. When you are sure that the current OSPFv2 LAN is operating correctly, you will connect the new LAN and verify that OSPF routes are being propagated for the new LAN. As branch office network administrator, you have full access to the IOS on routers R3 and R4. You only have read access to the enterprise LAN routers R1 and R2, using the username **BranchAdmin**, and the password **Branch1234**.

Instructor Note: The username **Admin**, password **Cisco1234** has full privilege level 15 on routers R1 and R2.

Instructions

Part 1: Verify the existing OSPFv2 network operation.

The following commands will help you find the information needed to answer the questions:

```
show ip interface brief
show ip route
show ip route ospf
show ip ospf neighbor
show ip protocols
show ip ospf
show ip ospf interface
```

Step 1: Verify OSPFv2 operation.

Wait until STP has converged on the network. You can click the Packet Tracer Fast Forward Time button to speed up the process. Continue only when all link lights are green.

- Log into router **R1** using the username **BranchAdmin** and the password **Branch1234**. Execute the **show ip route** command.

```
R1# show ip route
--- output omitted ---
```

```
Gateway of last resort is 172.16.3.2 to network 0.0.0.0
```

```

    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:02:18, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O       192.168.1.0/24 [110/65] via 192.168.10.6, 00:02:18, Serial0/0/1
    192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 172.16.3.2, 00:02:18, Serial0/0/0
                                   [110/128] via 192.168.10.6, 00:02:18, Serial0/0/1
```

```

O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:02:18, Serial0/0/0
R1#
R1#ena
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 172.16.3.2 to network 0.0.0.0

    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:07:58, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O       192.168.1.0/24 [110/65] via 192.168.10.6, 00:07:58, Serial0/0/1
        192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 192.168.10.6, 00:07:58, Serial0/0/1
        [110/128] via 172.16.3.2, 00:07:58, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:07:58, Serial0/0/0

R1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
2.2.2.2        0     FULL/  -        00:00:34    172.16.3.2     Serial0/0/0
3.3.3.3        0     FULL/  -        00:00:38    192.168.10.6   Serial0/0/1
R1#

```

แสดงตาราง routing ของ R1 ที่ได้รับ default route ผ่าน OSPF (O*E2) จาก R2 และแสดงสถานะ OSPF neighbor ที่เชื่อมกับ R2 และ R3 สำเร็จเป็น FULL ทั้งคู่

How did router **R1** receive the default route?

R1 ได้รับ default route ผ่านโปรโตคอล OSPF

From which router did **R1** receive the default route?

ได้รับจากเราเตอร์ R2

How can you filter the output of **show ip route** to show only the routes learned through OSPF?

สามารถใช้คำสั่ง **show ip route ospf** หรือ **show ip route | include O** เพื่กรองแสดงเฉพาะเส้นทางที่เรียนรู้ผ่าน OSPF

- b. Execute the **show ip ospf neighbor** command on **R1**.

Which routers have formed adjacencies with router **R1**?

R2 และ R3

What are the router IDs and state of the routers shown in the command output?

2.2.2.2 – สถานะ Full/- และ 3.3.3.3 – สถานะ Full/-

Are all of the adjacent routers shown in the output?

ใช่ แสดงครบทุกเราเตอร์ที่เป็น neighbor

- c. Using the command prompt on **PC1**, ping the address of the **ISP Router** shown in the Address Table. Is it successful? If not, do a **clear ospf process** command on the routers and repeat the ping command.

ผล ping 64.100.54.5 จาก PC1

```
C:\>ping 64.100.54.5

Pinging 64.100.54.5 with 32 bytes of data:

Reply from 64.100.54.5: bytes=32 time=10ms TTL=126
Reply from 64.100.54.5: bytes=32 time=5ms TTL=126
Reply from 64.100.54.5: bytes=32 time=5ms TTL=126
Reply from 64.100.54.5: bytes=32 time=9ms TTL=126

Ping statistics for 64.100.54.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 10ms, Average = 7ms

C:\>
```

Ping สำเร็จ

Step 1: Verify OSPFv2 operation on R2.

- a. Log into router **R2** using the username **BranchAdmin** and the password **Branch1234**. Execute the **show ip route** command. Verify that routes to all the networks in the topology are shown in the routing table.

```

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.3.1 to network 0.0.0.0

    64.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      C    64.100.54.4/30 is directly connected, GigabitEthernet0/1
      L    64.100.54.6/32 is directly connected, GigabitEthernet0/1
    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
      O    172.16.1.0/24 [110/65] via 172.16.3.1, 00:47:01, Serial0/0/0
      C    172.16.2.0/24 is directly connected, GigabitEthernet0/0
      L    172.16.2.1/32 is directly connected, GigabitEthernet0/0
      C    172.16.3.0/30 is directly connected, Serial0/0/0
      L    172.16.3.2/32 is directly connected, Serial0/0/0
      O    192.168.1.0/24 [110/65] via 192.168.10.10, 00:47:01, Serial0/0/1
    192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
      O    192.168.10.4/30 [110/128] via 192.168.10.10, 00:47:01, Serial0/0/1
           [110/128] via 172.16.3.1, 00:47:01, Serial0/0/0
      C    192.168.10.8/30 is directly connected, Serial0/0/1
      L    192.168.10.9/32 is directly connected, Serial0/0/1
      S*   0.0.0.0/0 [1/0] via 172.16.3.1
           [1/0] via 64.100.54.5

R2#

```

แสดง routing table ของ R2 รวมถึง static default route (S*) ที่ตั้งไว้เอง ซึ่งชี้ไปยัง ISP ผ่าน 2 เส้นทาง (172.16.3.1 และ 64.100.54.5)

How did router R2 learn the default route to the ISP?

It was statically configured by the administrator.

- b. Enter the **show ip ospf interface g0/0** on router **R2**.

What type of OSPF network is attached to this interface?

BROADCAST

Are OSPF hello packets being sent out this interface? Explain.

No. The interface is configured as a passive interface in OSPF.

```

R2#show ip ospf interface g0/0

GigabitEthernet0/0 is up, line protocol is up
Internet address is 172.16.2.1/24, Area 0
Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 2.2.2.2, Interface address 172.16.2.1
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  No Hellos (Passive interface)
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
R2#

```

แสดงรายละเอียด OSPF ของ interface G0/0 บน R2 ยืนยันว่า Network Type เป็น BROADCAST และไม่มี การส่ง Hello packet ออกทาง interface นี้ เนื่องจากถูกตั้งเป็น Passive Interface

- c. Using the command prompt on **PC2**, ping the S0/0/1 address on router **R3**.

Is it successful?

```

C:\>
C:\>
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=9ms TTL=254
Reply from 192.168.10.10: bytes=32 time=1ms TTL=254
Reply from 192.168.10.10: bytes=32 time=1ms TTL=254
Reply from 192.168.10.10: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 9ms, Average = 3ms
C:\>

```

Ping สำเร็จ

Step 2: Verify OSPFv2 operation on R3.

- a. Execute the **show ip protocols** command on router R3.

Router R3 is routing for which networks?

192.168.1.0/24, 192.168.10.4/30, and 192.168.10.8/30

```
R3
00:34:15: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0/1 from LOADING to FULL, Loading Done
00:34:15: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/0/0 from LOADING to FULL, Loading Done

R3>enable
R3#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.1.0 0.0.0.255 area 0
    192.168.10.4 0.0.0.3 area 0
    192.168.10.8 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1           110          00:20:00
    2.2.2.2           110          00:15:36
    3.3.3.3           110          00:20:00
  Distance: (default is 110)

R3#
```

แสดงข้อมูล OSPF process ของ R3 ได้แก่ Router ID, เครือข่ายที่ประกาศเข้า OSPF (192.168.1.0/24, 192.168.10.4/30, 192.168.10.8/30) และ Passive Interface ที่ตั้งไว้ (G0/0)

- b. Execute the **show ip ospf neighbor detail** command on router **R3**.

```
R3#show ip ospf neighbor detail
Neighbor 1.1.1.1, interface address 192.168.10.5
  In the area 0 via interface Serial0/0/0
  Neighbor priority is 0, State is FULL, 7 state changes
  DR is 0.0.0.0 BDR is 0.0.0.0
  Options is 0x00
  Dead timer due in 00:00:30
  Neighbor is up for 00:50:41
  Index 1/1, retransmission queue length 1, number of retransmission 603
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 1, maximum is 3
  Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 2.2.2.2, interface address 192.168.10.9
  In the area 0 via interface Serial0/0/1
  Neighbor priority is 0, State is FULL, 7 state changes
  DR is 0.0.0.0 BDR is 0.0.0.0
  Options is 0x00
  Dead timer due in 00:00:30
  Neighbor is up for 00:50:41
  Index 2/2, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 1
  Last retransmission scan time is 0 msec, maximum is 0 msec
--More-- |
```

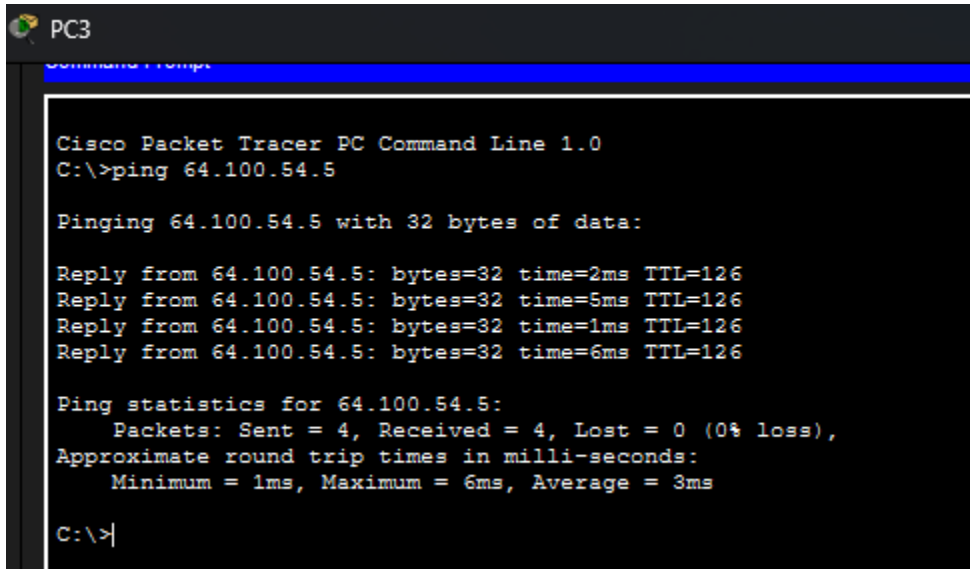
แสดงรายละเอียด Neighbor ของ R3 ยืนยันว่า Neighbor priority ของ router ข้างเคียง (R1, R2) เท่ากับ 0 เนื่องจากเป็นการเชื่อมต่อแบบ Point-to-Point ผ่าน Serial link

What is the neighbor priority shown for the OSPF neighbor routers?

0

- c. Using the command prompt on **PC3**, ping the address of the **ISP Router** shown in the Address Table.

Is it successful?



```
PC3
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.54.5

Pinging 64.100.54.5 with 32 bytes of data:

Reply from 64.100.54.5: bytes=32 time=2ms TTL=126
Reply from 64.100.54.5: bytes=32 time=5ms TTL=126
Reply from 64.100.54.5: bytes=32 time=1ms TTL=126
Reply from 64.100.54.5: bytes=32 time=6ms TTL=126

Ping statistics for 64.100.54.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>|
```

Ping สำเร็จ

Part 2: Add the new Branch Office LAN to the OSPFv2 network.

You will now add the pre-configured Branch Office LAN to the OSPFv2 network.

Step 1: Verify the OSPFv2 configuration on router R4.

Execute a **show run | begin router ospf** command on router **R4**. Verify that the network statements are present for the networks that are configured on the router.

Which interface is configured to not send OSPF update packets?

```

R4>
R4>enable
R4#show run | begin router ospf
router ospf 1
  router-id 4.4.4.4
  log-adjacency-changes
  passive-interface GigabitEthernet0/0/1
  network 192.168.1.0 0.0.0.255 area 0
  network 192.168.11.0 0.0.0.255 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end
R4#

```

แสดง configuration OSPF ของ R4 ยืนยันว่า interface GigabitEthernet0/0/1 ถูกตั้งเป็น Passive Interface (ไม่ส่ง OSPF update packet)

Interface GigabitEthernet0/0/1

Step 2: Connect the Branch Office router R4 to the OSPFv2 network.

- Using the correct Ethernet cable, connect the G0/0/0 interface on router **R4** to the G0/1 interface on switch **S3**. Use the **show ip ospf neighbor** command to verify that router **R4** is now adjacent with router **R3**.

What state is displayed for router **R3**?

```

R4#
R4#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/BDR	00:00:36	192.168.1.1	GigabitEthernet0/0/0

```

R4#

```

แสดงสถานะ Adjacency ของ R4 กับ R3 (3.3.3.3) เป็น FULL/BDR ผ่าน interface GigabitEthernet0/0/0
ยืนยันว่าการเชื่อมต่อ Branch Office LAN เข้ากับเครือข่าย OSPF หลักสำเร็จ

FULL/DR

- b. Using the **show ip ospf neighbor** command on **R3**, determine the state of router **R4**. There may be a delay while OSPF converges.

Why is the state of router R4 different than the state of R1 and R2?

```
R3>
R3>
R3>show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:36	192.168.10.5	Serial10/0/0
4.4.4.4	1	FULL/DR	00:00:36	192.168.1.2	GigabitEthernet0/0
2.2.2.2	0	FULL/ -	00:00:36	192.168.10.9	Serial10/0/1

```
R3>
```

แสดงสถานะ Adjacency ของ R3 กับ neighbor ทั้งหมด ได้แก่ R1, R2 (state FULL ผ่าน Serial) และ R4 (state FULL/DR ผ่าน GigabitEthernet0/0) ยืนยันว่า R3 ทำหน้าที่เป็น Designated Router (DR) บนเครือข่าย Broadcast segment ที่เชื่อมกับ R4

Because the OSPF network type between R1 and R2 is Point-to-Point, there is no OSPF election. R4 is on the same Ethernet network segment as router R3, so the OSPF network type is Broadcast and there is an OSPF election. When more than one router is located on a multiaccess network segment, only one router, the DR, sends OSPFv2 updates. A second router, in this case R4, becomes the Backup Designated Router and can take over if the DR router fails.

- c. Using the command prompt on Laptop, ping the address of PC2.

ตั้ง DHCP pool บน R4

```
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip dhcp excluded-address 192.168.11.1
R4(config)#ip dhcp pool BRANCH-LAN
R4(dhcp-config)# network 192.168.11.0 255.255.255.0
R4(dhcp-config)# default-router 192.168.11.1
R4(dhcp-config)#
```

Is it successful?

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.16.2.2: bytes=32 time=12ms TTL=125
Reply from 172.16.2.2: bytes=32 time=10ms TTL=125
Reply from 172.16.2.2: bytes=32 time=8ms TTL=125

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 12ms, Average = 10ms

C:\>
```

Ping สำเร็จ

Yes